

Substitution method



1

a $x = -8, y = 8$

c $x = -8, y = -3$

e $x = -3, y = 5$

g $x = 9, y = 5$

i $x = 1, y = 6$

k $x = -5, y = -8$

m $x = -\frac{4}{15}, y = \frac{13}{3}$

b $x = -5, y = 3$

d $x = -6, y = -3$

f $x = 6, y = 4$

h $x = -2, y = 9$

j $x = 8, y = 4$

l $x = 8, y = 1$

n $x = 7, y = 4$

2

a $x = \frac{4 + 7y}{3}$

c Dependent

b Infinitely many solutions.

3

a $y = -\frac{64}{3} - \frac{5x}{3}$

b $x = -8$

c $y = -8$

4

a $x = 7, y = 4$

b $x = -6, y = 5$

c $x = 6, y = -5$

d $x = 2, y = 3$

e $f = \frac{2}{5}, g = \frac{6}{5}$

f $q = \frac{4}{5}, p = \frac{1}{8}$

g $a = \frac{3}{2}, b = \frac{5}{2}$

h $v = \frac{2}{5}, t = \frac{5}{4}$

5

a $x = 4, y = 2$

b Yes

c Yes

Applications

6

a $x = y + 10$

b $x + y = 28$

c $y = 9$

d $x = 19$

7

a $y = x + 6$

b $x + y = 22$

c $x = 8$

d $y = 14$

8

a $y = 10x, y = 7x + 12$

b $x = 4$

c $y = 40$

9

a $y = 5x, y = 9x - 32$

c $y = 40$



b $x = 8$

10

a $x + y = 3000, 0.06x + 0.08y = 204$

b $x = \$1800$ and $y = \$1200$

11

a $2f + 3g = 10.08$ and $5f + 7g = \$24.08$

b $f = \$1.68$ and $g = \$2.24$

12

a $y = 4x - 190\,000, y = x + 560\,000$

b $x = 250\,000$ and $y = 810\,000$

13

a $x + 10y = 105$ and $x + 15y = 155$

b $x = 5, y = 10$

c \$255

14 Each racket is \$500 and each ball is \$50.

15 15 pieces of 10c coins and 45 pieces of 5c coins

16 35 hotdogs and 52 sodas.